



Lahore Grammar School
Johar Town Senior Boys Campus

INFINITESIMA

Study Guide



INFINITESIMA

11 . 12 . 13
SEPTEMBER





Round 1:

Delegate Count: 2-3

Duration: 45 minutes

To start off the category, teams will take a traditional written test with pen and paper. Each team answers 10 multiple-choice questions and 10 short-answer questions spanning the full breadth of competition mathematics, from geometry to combinatorics. Multiple-choice questions carry negative marking, so guessing carries real risk.

Note: This is an accumulation round.

Round 2:

Delegate Count: 2-3

Duration: 1 hour 30 minutes

This round tests self-awareness as much as mathematical ability. Each team is shown a menu of problems, but the problems themselves stay hidden, every problem is labelled only with a difficulty tier (Bronze, Silver and Gold). Teams get a short time to study this menu and lock in their menu selection, highlighting the problems they intend to attempt, before any problem statement is revealed. Once every team has submitted their choice, the actual questions are unveiled and the solving period begins.

Note: This is an elimination round.





Round 3:

Delegate Count: 1-2

Duration: 3 hours

This round takes place in an auction room, where every bid can be the difference between victory and defeat. Every team starts this round with the same budget. Problems are grouped by mathematical domain, and harder problems open at a higher starting bid. The round unfolds in waves; In each wave teams bid on the problems on offer, and the highest bidder wins the right to attempt that problem. The pool of available problems is limited, so bidding aggressively early can leave a team short of funds, and therefore short of problems, later in the round. After the auction closes, teams get to solve whatever problems they have won.

Note: This is an elimination round.

Round 4: (Final Round)

Delegate Count: 1-2

Duration: 2 hours

The surviving teams are split randomly into two groups. Within each group, teams compete head-to-head in a live jeopardy-style game: on your turn, you choose a square from the grid, selecting both the mathematical domain and the difficulty level of the question you want to face. Once a question is chosen, the team has a limited time to solve it. Only correct answers within the given time will be awarded points.





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Note:

- Winner and Runner Up will be decided by culmination of points.
- Final decisions are up to the judges/category heads.
- Calculators will be allowed in certain rounds
- Category heads have the complete authority to change the course of the competition at any time

Code of conduct:

all participants are expected to;

- Demonstrate good sportsmanship.
- Respect fellow competitors and event officials.
- Maintain a clean and organized workspace.
- Accept the judges' decisions respectfully.
- Use of artificial intelligence is strictly prohibited
- All rules and instructions laid out by the category heads have to be followed respectfully
- Any form of cheating, intentional interference, unsafe behaviour, or misconduct may result in penalties or disqualification

Items to bring:

- Calculators (Non programmable)
- Basic stationery

